

# Prophylactic Use of Bevacizumab to Avoid Anterior Segment Neovascularization Following Proton Therapy for Uveal Melanoma

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- **PURPOSE:** To investigate whether the prophylactic use of bevacizumab reduces the rate of rubeosis after proton therapy for uveal melanoma and improves the possibility to treat ischemic, reapplicated retina with laser photocoagulation.

- **DESIGN:** Comparative retrospective case series.

- **METHODS:** Uveal melanoma patients with ischemic retinal detachment and treated with proton therapy were included in this institutional study. Twenty-four eyes received prophylactic intravitreal bevacizumab injections and were compared with a control group of 44 eyes without bevacizumab treatment. Bevacizumab injections were performed at the time of tantalum clip insertion and were repeated every 2 months during 6 months, and every 3 months thereafter. Ultra-widefield angiography allowed determination of the extent of retinal ischemia, which was treated with laser photocoagulation after retinal reapplication. Main outcome measures were the time to rubeosis, the time to retinal reattachment, and the time to laser photocoagulation of ischemic retina.

- **RESULTS:** Baseline characteristics were balanced between the groups, except for thicker tumors and larger retinal detachments in the bevacizumab group, potentially to the disadvantage of the study group. Nevertheless, bevacizumab prophylaxis significantly reduced the rate of iris rubeosis from 36% to 4% (log-rank test  $P = .02$ ) and tended to shorten the time to retinal reapplication until laser photocoagulation of the nonperfusion areas could be performed.

- **CONCLUSIONS:** Prophylactic intravitreal bevacizumab in patients treated with proton therapy for uveal melanoma with ischemic retinal detachment prevented anterior segment neovascularization, until laser photocoagulation to the reappplied retina could be performed. (Am J

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**P**ROTON THERAPY FOR UVEAL MELANOMA HAS shown a very high success rate in terms of local tumor control (94%–99% after 5 years)<sup>1–3</sup> and eye retention probability (89%–91% after 5 years).<sup>4–6</sup> However, some irradiated eyes still undergo secondary enucleation owing to radiation-induced complications. Relative risk analyses have indicated that tumor height,<sup>2,4,5,7,8</sup> tumor diameter,<sup>2,7</sup> ciliary body involvement,<sup>5</sup> and proximity to the optic disc<sup>4</sup> or fovea<sup>5</sup> are significant risk factors for enucleation. Neovascular glaucoma has been cited as a primary cause of enucleation.<sup>4–6,9</sup> Rates of neovascular glaucoma following proton therapy for uveal melanoma vary widely in the literature (7%–47%).<sup>2,9,10</sup> Neovascular glaucoma has been associated with tumor size,<sup>4,7,9,11</sup> proximity to the optic nerve<sup>4,12</sup> or macula,<sup>5,8</sup> uveal involvement anterior to the equator,<sup>5,7</sup> and the presence of a serous retinal detachment.<sup>4,9</sup> Our group has previously demonstrated that Baerveldt shunt implantation is a successful treatment option for neovascular glaucoma in cases of anterior uveal melanoma treated with proton therapy, which negated the need for enucleation owing to neovascular glaucoma.<sup>13</sup> However, the ideal situation would be to prevent iris neovascularization and the development of neovascular glaucoma and avoid this secondary surgical intervention with its associated risks.

The advent of intravitreal anti-vascular endothelial growth factor (VEGF) injections has provided an efficacious treatment for anterior segment neovascularization.<sup>14</sup> However, it does not always prevent secondary enucleation when applied in florid neovascularization. The use of prophylactic intravitreal anti-VEGF to inhibit iris neovascularization in uveal melanoma eyes treated with proton therapy has not been previously reported. The aim of this study was to examine whether the prophylactic use of bevacizumab in eyes with a high risk profile for neovascular glaucoma following proton therapy may reduce the rate of anterior segment neovascularization. In addition, we investigated whether the use of anti-VEGF increased the rate of retinal reapplication and reduced the time until photocoagulation of ischemic retina became possible.

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